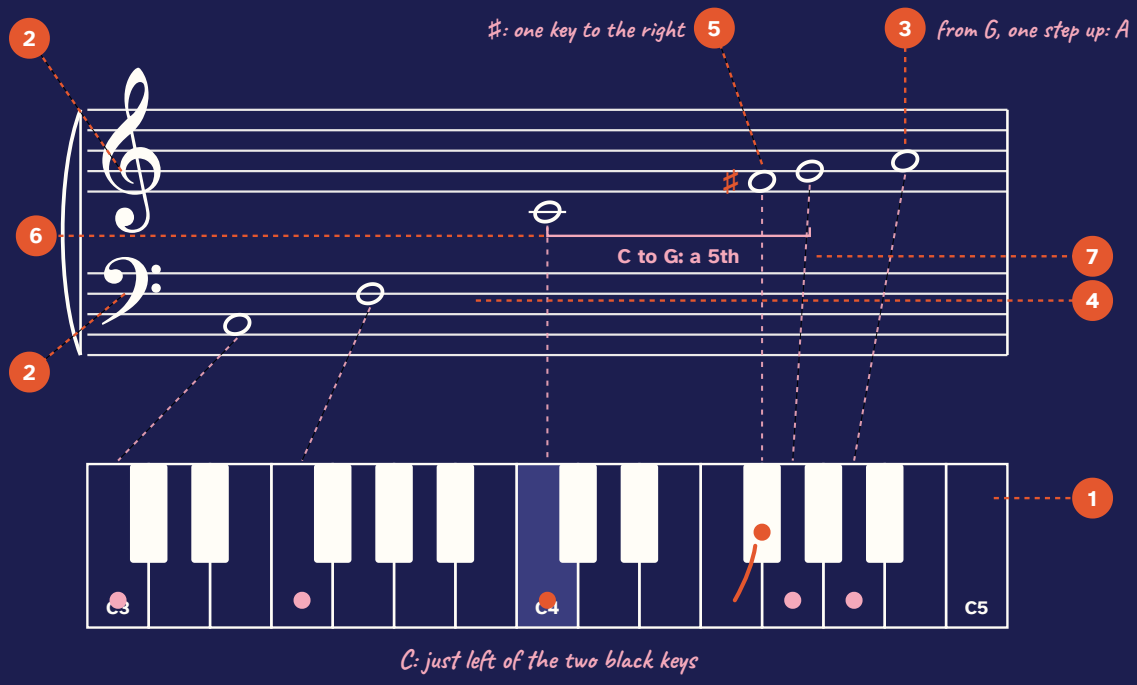
PART I · WHICH NOTE

Pitch

High or low? That is the first question every note answers. Start with the keys under your fingers, find where they live on the staff, see how sharps and flats move them, then measure how far apart two notes are. Everything in this part is in this one picture.



1 **The keyboard** p. 22
two and three black keys; C is left of the two

2 **The staff and the clefs** p. 25
the treble clef names G, the bass clef F

3 **Reading the notes** p. 28
find a landmark note, then step or skip from it

4 **The staff and the keyboard** p. 31
every line and space is one white key

5 **Sharps, flats and naturals** p. 34
a sharp moves one key right, a flat one key left

6 **Counting intervals** p. 37
count the letters, both ends: C to G is a 5th

7 **Major, minor and perfect** p. 40
count the semitones: 7 make a perfect 5th

★ **Tips: when a note reads wrong** p. 43

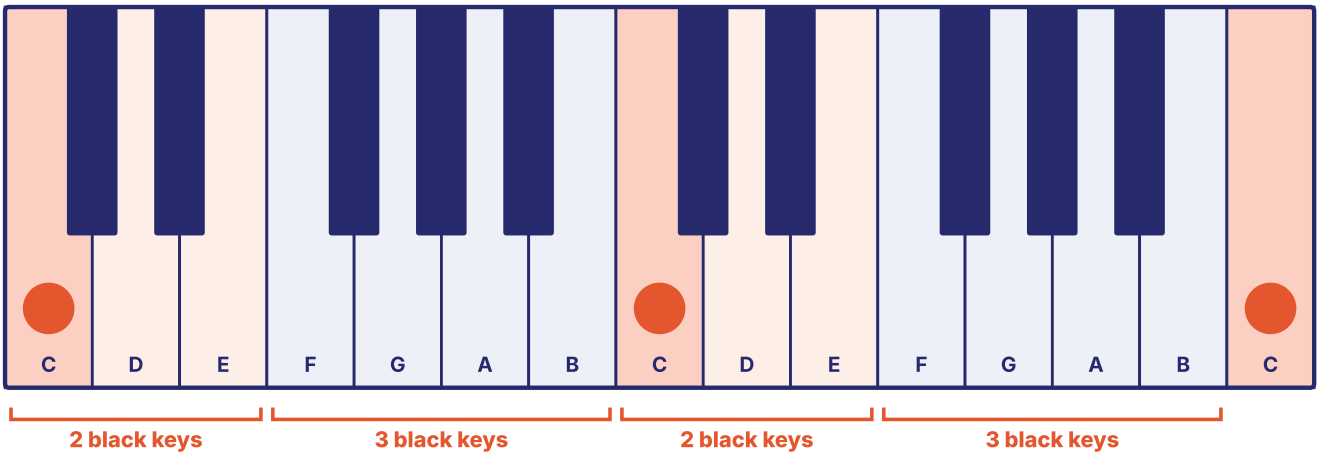
PART I · PITCH · 1

The keyboard

Keys to the **left** sound **lower**; keys to the **right** sound **higher**. The black keys come in groups of **two** and **three**, and that pattern repeats all the way along: it is how you find every note.

THE SECRET

Find the two black keys: C is always just to their left.



Every C sits just left of the two black keys

The white keys

Seven letters, **A B C D E F G**, then the letters start again, like the days of the week. **C D E** sit around each group of two black keys (C just left of it), **F G A B** around each group of three (F just left of it): the two tints on the picture.

Middle C and octaves

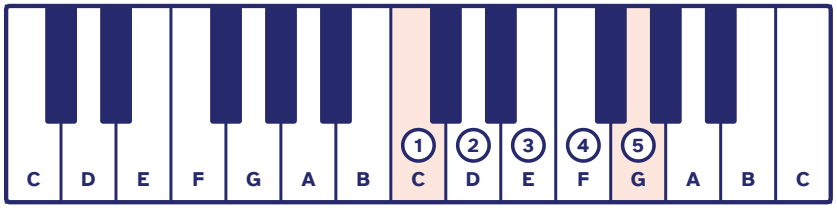
Middle C is the C nearest the middle of the keyboard. From one C to the next is an **octave**: eight white keys, counting both. Octaves are numbered from the bottom: middle C is **C4**, the C below it C3, the C above C5; every key up to the next B shares the number (D4, E4 … B4).

TERM	MEANING	FIRST MET
pitch	how high or low a sound is	Level 0
register	a part of the keyboard: low, middle or high	Level 0
semitone	from one key to the very next key, black or white	Level 0
tone	two semitones	Level 0
octave	from one note to the next with the same name (8 white keys, 12 semitones)	Level 0

PART I · PITCH · 1 · CONTINUED

EXAMPLE 1

Find G4 on the keyboard

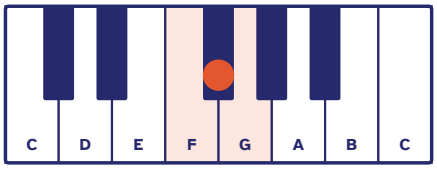


1. **Find middle C (C4):** the C just left of the two black keys nearest the middle.
2. **Count up the letters**, one white key each: C D E F G is five keys.
3. **Same octave**, so the number stays 4.

The key: **G4**. One octave higher, the next G, is G5.

EXAMPLE 2

Find F♯4

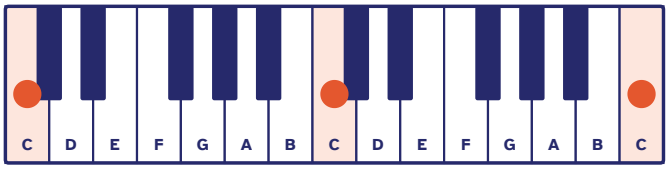


1. **Find F4:** the white key just left of the group of three black keys above middle C.
2. **Sharp means up:** move one key to the right.
3. **The very next key** is black: that is F♯.

F♯4 is the **black key** just right of F4, the first of the group of three.

EXAMPLE 3

Find C3, C4 and C5



1. **Find middle C (C4):** just left of the two black keys in the middle.
2. **Count seven white keys down:** C3.
3. **Count seven white keys up:** C5.

Every C sits just left of a group of **two** black keys, an octave apart.

TRY IT · 10 SECONDS



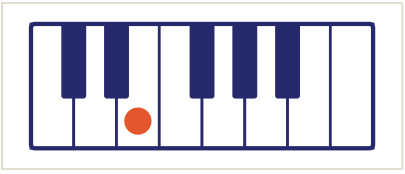
Play every C on the piano.

Start at the bottom and work up, saying “C” each time: left of the two black keys, every time. Then every F: left of the three.

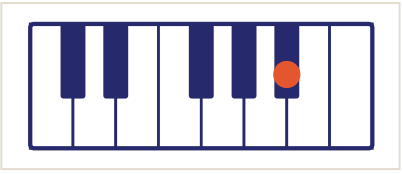
PART I · PITCH · 1 · CONTINUED

GAME

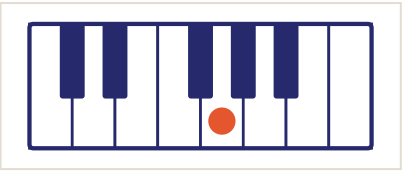
Name the key



1



2



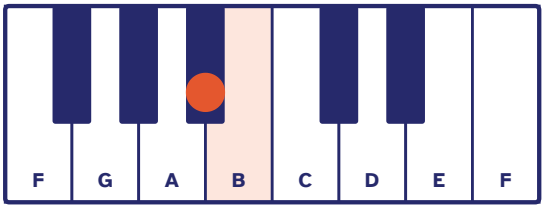
3

Each keyboard starts on C. Use the groups of black keys to name the red key.

Answer: 1 E (right of the two) · 2 the black key between A and B: A♯ or B♭ · 3 G (in the group of three).

EXAMPLE 4

Find B♭3

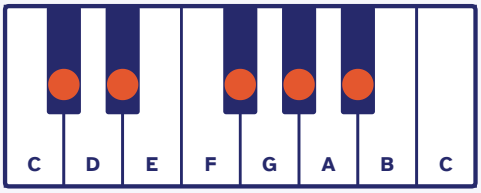


1. **Find middle C (C4)**, then the white key just below it: **B3**.
2. **Flat means down:** one key to the left.
3. **The very next key** is black: the top key of the group of three.

B♭3 is the **top black key** of the group of three just below middle C.

YOU KNOW THIS

Amazing Grace, on the black keys only



The whole tune fits on the **five black keys**: D♭ G♭ B♭ G♭ B♭ A♭ G♭ E♭ D♭. The groups of two and three are a little melody machine.

Try it. Start on the lower key of a pair of black keys and find the tune by ear.

THINK OF IT The black keys are like the little bumps on F and J of a computer keyboard: feel them and you know where you are without looking.

TRAP

× Calling every black key a sharp: each one has a flat name too (p. 34).

SEE ALSO

The staff and the clefs p. 25 · The staff and the keyboard p. 31

PRACTISE IT

Workbook Pitch 1 · Online: Library › Keyboard

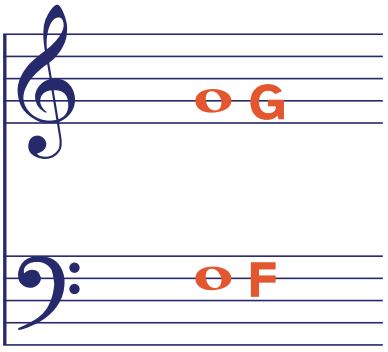
PART I · PITCH · 2

The staff and the clefs

Music lives on a **staff** (or stave): five lines and four spaces, counted from the **bottom**. Higher on the staff means higher in sound. The **clef** at the start is the code: it says which note each line and space stands for.

THE SECRET

The treble clef curls round G; the bass clef's two dots hug F.



The treble clef curls round G · the bass clef's dots hug F

CLEF	NAME	HOW IT SHOWS ITS NOTE	FIRST MET
𝄞	**treble clef** (G clef)	curls round the line for G (line 2)	Level 0
𝄢	**bass clef** (F clef)	its two dots sit either side of the line for F (line 4)	Level 0



Alto clef: middle C on line 3. Used by the viola.

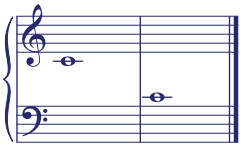


Tenor clef: middle C on line 4. Cello, bassoon and trombone, up high.

Other clefs. C clefs: the line through the centre of the clef is middle C. Piano music never uses them, but pianists meet them in theory exams and when reading the viola or cello part in chamber music or a score (Level 2).

The grand staff

Piano music uses two staves joined by a brace: the treble staff, usually for the right hand, above the bass staff, usually for the left. **Middle C** sits between them on a short line of its own, and can be written in either staff.

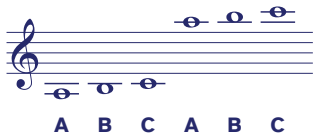


Middle C in the treble staff (left) and in the bass staff (right): the same key.

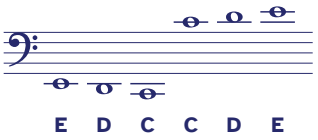
Ledger lines

Notes above or below the staff sit on short extra lines called **ledger lines**, spaced like the staff lines.

PART I · PITCH · 2 · CONTINUED



Treble: A, B, C below; A, B, C above.



Bass: E, D, C below; C, D, E above.

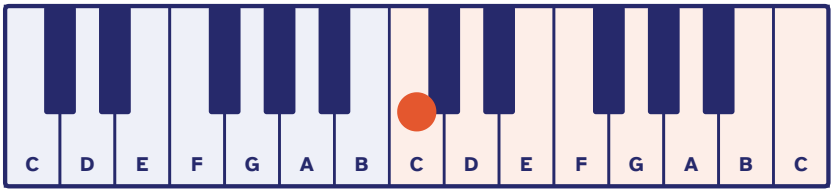
Octave signs

SIGN	MEANING
8va	an octave higher, as far as the dotted line
8vb	an octave lower
loco	back where it is written

First met: Level ½. Octave signs save writing many ledger lines: a high passage stays on the staff, with 8va above it.

EXAMPLE 1

Which staff does this note belong to?



1. **Below middle C** (blue): the bass staff, usually the left hand.
2. **Above middle C** (pink): the treble staff, usually the right hand.
3. **Middle C itself** (the red dot): either staff, whichever hand plays it.

Low keys: **bass clef**. High keys: **treble clef**. The two staves meet at middle C.

GAME

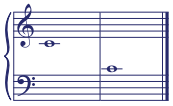
Treble or bass?

1. a note two keys above middle C
2. the lowest key on the piano
3. middle C
4. a high, high note

Which staff would each note be written on?

Answer: 1 treble · 2 bass · 3 either: it sits between the two · 4 treble, with ledger lines or 8va.

TRY IT · 10 SECONDS



Point, then play.

Point to the line the treble clef curls round and play that G. Point to the line between the bass clef's dots and play that F. Middle C sits between them.

PART I · PITCH · 2 · CONTINUED

EXAMPLE 2

Read a high note on ledger lines



1. **The top line** of the treble staff is F.
2. **Step up:** G sits on top of the staff, A on the first ledger line.
3. **Keep stepping:** B above it, C on the second ledger line.

Two ledger lines above the treble staff: **C6**, two octaves above middle C.

EXAMPLE 3

One line, two notes
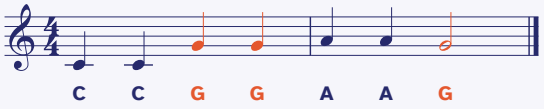


1. **Treble clef:** the bottom line is E.
2. **Bass clef:** the same bottom line is G.
3. **Same line, new note:** the clef decides.

Always read the **clef** first: it tells you what every line means.

YOU KNOW THIS

Twinkle, Twinkle, Little Star



The treble clef curls round **line 2: G**, the very note Twinkle leaps up to. Middle C hangs on its own ledger line below.

Try it. Point to the G line, then play C C G G A A G.

THINK OF IT A clef is a pair of glasses: the same five lines read as different notes, depending on which pair you put on.

TRAP

× Reading bass-clef notes as if they were treble: check the clef on every line.

SEE ALSO
Reading the notes p. 28 · The staff and the keyboard p. 31

PRACTISE IT
Workbook Pitch 2 · Online: Library › Staff and clefs

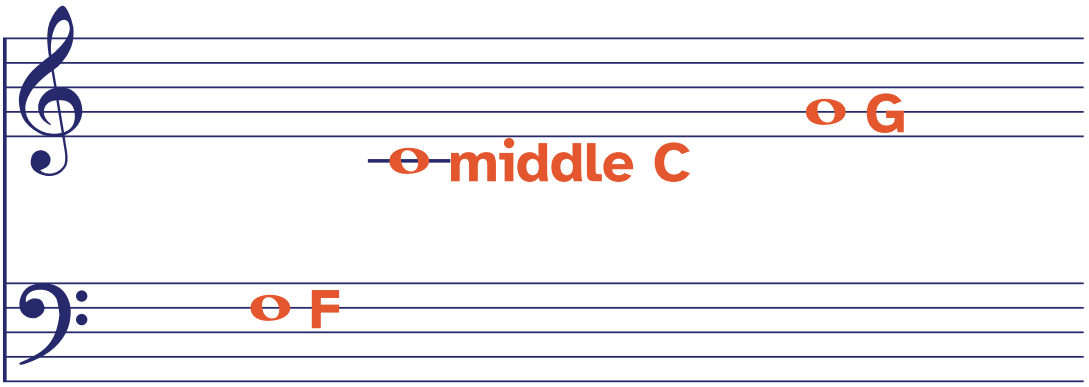
PART I · PITCH · 3

Reading the notes

Each line and space has one letter name. Learn them in two ways: by the patterns below, and by a few **landmark notes** you can find instantly and count from.

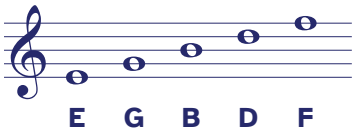
THE SECRET

Learn three landmark notes, then step or skip from them.



Find a landmark, then count

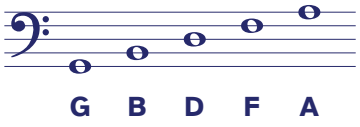
Find the three red **landmark notes** first: bass F, middle C, treble G. Every other note is a **step** (line to space) or a **skip** (line to line) away from one of them.



Treble clef: the lines



Treble clef: the spaces



Bass clef: the lines



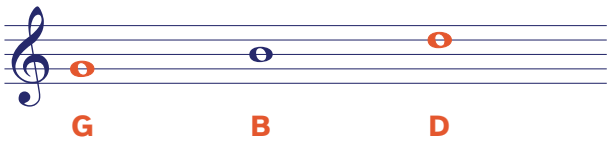
Bass clef: the spaces

STAFF	NOTES, BOTTOM TO TOP	TO REMEMBER
Treble lines	E G B D F	*Every Good Boy Deserves Fun*
Treble spaces	F A C E	spells *FACE*
Bass lines	G B D F A	*Great Big Dragon Flies Away*
Bass spaces	A C E G	*All Cows Eat Grass*

PART I · PITCH · 3 · CONTINUED

EXAMPLE 1

Read a note you are not sure of



1. **The clef:** treble.
2. **A landmark:** the G the clef curls round, on line 2.
3. **Skip up** line to line, one letter jumped each time: G, B, D.

The note on line 4 is **D**.

GAME

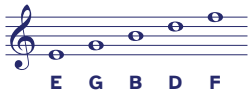
Fill the gap

1	2	3	4
E G _ D F	F A _ E	G B D _ A	A C E _

Treble lines, treble spaces, bass lines, bass spaces: one letter is missing from each.

Answer: 1 B · 2 C · 3 F · 4 G.

TRY IT · 10 SECONDS

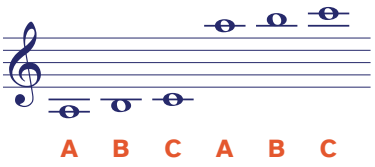


Say the treble lines up and down.

"E G B D F", then "F D B G E", out loud, as fast as you can without a slip. Then the spaces: "F A C E".

EXAMPLE 2

Read a ledger-line note



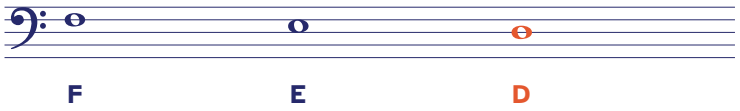
1. **Start at the staff edge:** the top line of the treble staff is F.
2. **Step up** one note at a time: G (space), A (first ledger line).
3. **Below the staff** count down the same way: D, C, B, A.

Ledger lines are the staff carried on: count by steps from the nearest line.

PART I · PITCH · 3 · CONTINUED

EXAMPLE 3

Read a bass-clef note from F

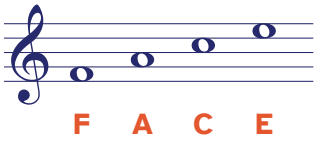


1. **The bass clef's dots** sit either side of line 4: that line is F.
2. **Step down** to the space below: E.
3. **One more step** to line 3: D.

Bass line 3 is **D**: two steps down from the clef's F.

EXAMPLE 4

Read a treble space note
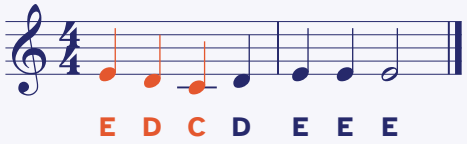


1. **Start from the clef's G** on line 2.
2. **Step up** to the space above it: A.
3. **Skip up** to the next space: C.

Space 3 of the treble staff is **C**: one of the landmark notes.

YOU KNOW THIS

Mary Had a Little Lamb



E D C: **line, space, ledger line**, one step down each time. Read the shape and the tune walks down with it.

Try it. Read the notes aloud before you play them.

THINK OF IT Landmark notes are like lamp posts on a dark road: walk from one to the next and you are never lost.

TRAP

- × Naming bass-clef notes with treble names: in the bass, the bottom line is G, not E.
- × Counting ledger lines from the wrong end: start from the staff and count outwards.

SEE ALSO

The staff and the clefs p. 25 · The staff and the keyboard p. 31

PRACTISE IT

Workbook Pitch 2 · Online: Library › Staff and clefs

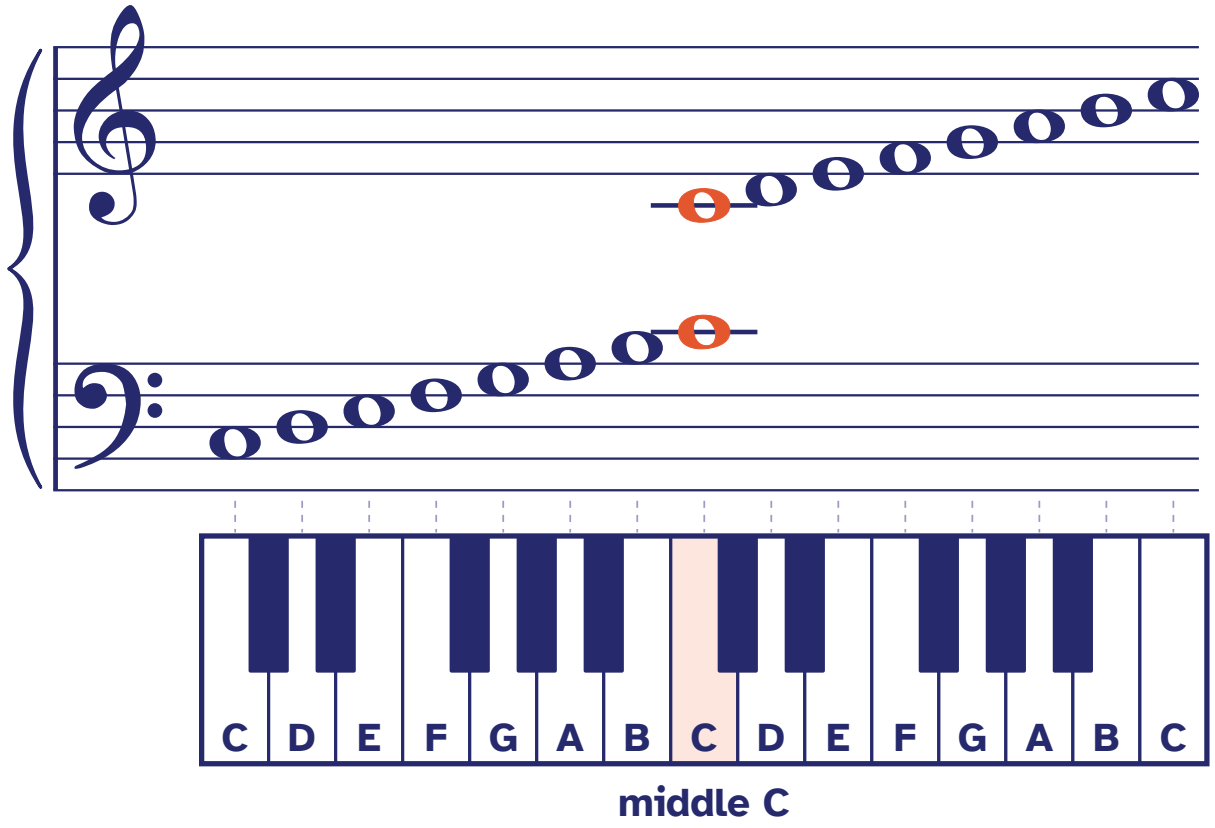
PART I · PITCH · 4

The staff and the keyboard

Every line and space of the grand staff is one white key. Read from left to right, the notes climb the staff exactly as the keys climb the keyboard: two octaves from C3 to C5, with middle C in red where the two staves meet.

THE SECRET

Every line and every space is one white key, in order.



What the map shows

Each step up the staff (line to space, space to line) is the next **white** key. The black keys have no line or space of their own: they are written with a sharp or a flat on a white key's line or space (p. 34).

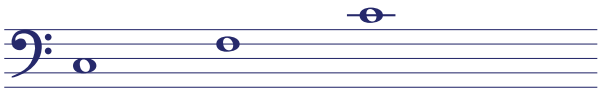
Two ways to write one key

Middle C can sit on a ledger line below the treble staff or above the bass staff. Both mean the same key; the music chooses the staff of the hand that plays it.

PART I · PITCH · 4 · CONTINUED

EXAMPLE 1

From the page to the key



C F middle C

1. **Name the note on the staff:** line 4 of the bass staff, between the clef's dots, is F.
2. **Count down from middle C:** C B A G F, the fifth white key down.
3. **Play it** with the left hand: that is F3.

The bass-clef F is **F3**, five white keys below middle C.

GAME

How many white keys?

1 middle C up to A

2 middle C down to E

3 G up to the next G

Count both ends. The staff climbs the same way: one line or space per white key.

Answer: 1 six (C D E F G A) · 2 six (C B A G F E) · 3 eight: an octave.

TRY IT · 10 SECONDS

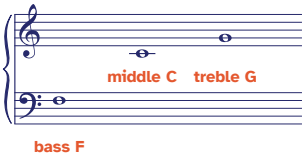


Play middle C, then walk up.

Say each white key as you play it, C D E F G A B C, and picture each one climbing the staff: line, space, line, space.

EXAMPLE 2

Landmarks either side of middle C



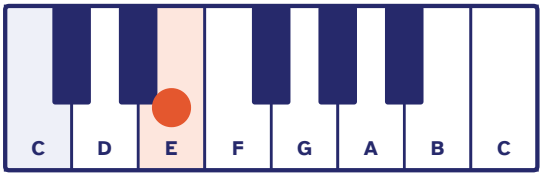
1. **Find middle C** on the map: between the two staves.
2. **Treble G** is five white keys above it.
3. **Bass F** is five white keys below it.

The landmarks sit at the same distance either side of middle C.

PART I · PITCH · 4 · CONTINUED

EXAMPLE 3

From the key to the page

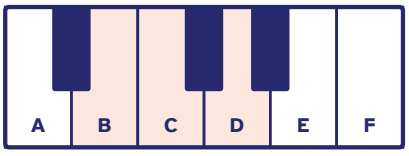


1. **The key:** E, two white keys above middle C.
2. **On the staff,** step up from middle C: D in the space below the staff, E on the bottom line.
3. **So the note sits** on the treble's bottom line.

E4 is the **bottom line** of the treble staff.

EXAMPLE 4

Where the two staves meet

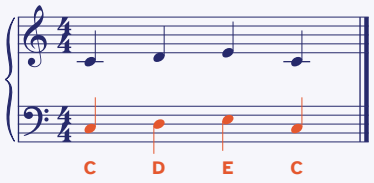


1. **The bass staff's top line** is A3.
2. **The treble staff's bottom line** is E4.
3. **The keys between,** B3 C4 D4, sit just outside each staff (C4 on its own ledger line).

B, C and D in the middle can be written in **either staff**: the hand that plays them chooses.

YOU KNOW THIS

Frère Jacques, in both hands



The left hand plays the **same letters**, C D E C, one octave lower, written in the bass clef.

Try it. Play it right hand, then left hand, then both together.

THINK OF IT The staff is a ladder laid on its side beside the keyboard: each rung, a line or a space, is the next white key.

TRAP

× Thinking the black keys have their own lines: they borrow a white key's line or space with a sharp or flat.

SEE ALSO
The keyboard p. 22 · Reading the notes p. 28

PRACTISE IT
Workbook Pitch 1–2 · Online: Library › Staff and clefs

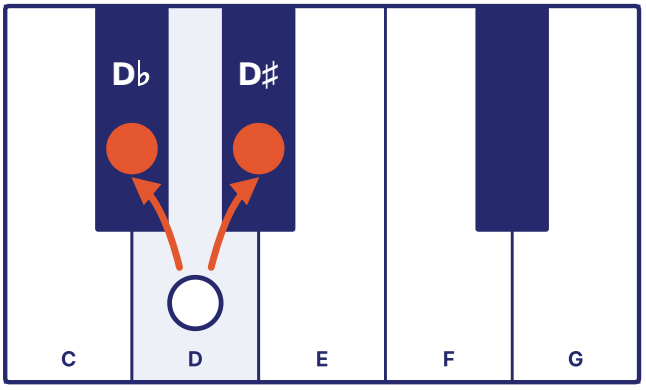
PART I · PITCH · 5

Sharps, flats and naturals

A sharp, flat or natural in front of a note (an **accidental**) nudges it, usually by a semitone. **Sharp**: one key right, usually black. **Flat**: one key left.

THE SECRET

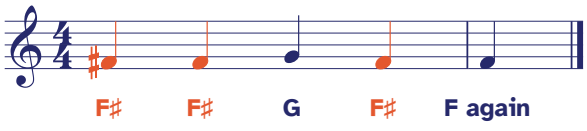
Sharp: the very next key to the right. Flat: the very next key to the left.



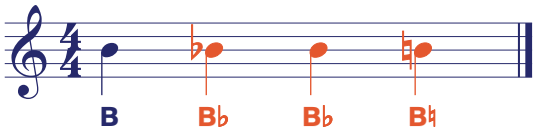
♯ one key right · ♭ one key left

SIGN	NAME	WHAT IT DOES	FIRST MET
♯	sharp	raises a note by a semitone	Level 0
♭	flat	lowers a note by a semitone	Level 0
♮	natural	cancels a sharp or flat	Level ½
𝄪	double sharp	raises by two semitones	Level 4
𝄫	double flat	lowers by two semitones	Level 4

Sharp till the bar line



The sharp still counts later in the bar; after the bar line, F is F again.



A natural cancels the flat for the rest of the bar.

Only that line or space. A sharp on the F in the first space does not change the F an octave higher.

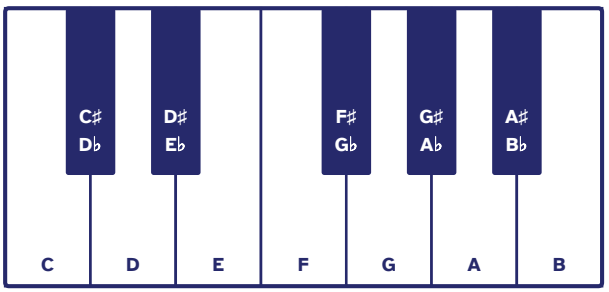
A tie carries it on. A note tied across the bar line keeps its accidental.

Reminders. A courtesy accidental, sometimes in brackets, just reminds you after a bar line.

PART I · PITCH · 5 · CONTINUED
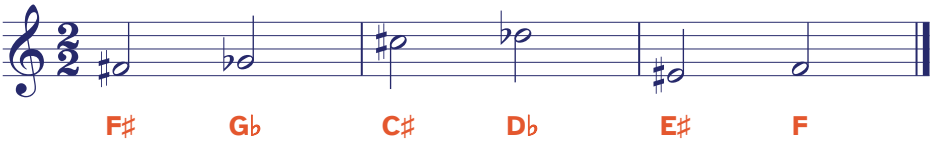
One key, two names

Every black key has two names: the key between F and G is both **F♯** and **G♭**. Notes that sound the same but are written differently are **enharmonic**. White keys can have sharp or flat names too: **E♯** is F, **C♭** is B.



EXAMPLE 1

One key, two names



1. **The first bar:** F♯, then G♭.
2. **Find them:** both are the black key between F and G.
3. **The last bar:** no black key sits above E, so E♯ is the white key F.

Same key, different names: these are **enharmonic** notes.

GAME

Which key is it?

1	2	3	4
C♯	E♭	B♯	F♭

Sharp: one key to the right. Flat: one key to the left. Which key do you press?

Answer: 1 the black key right of C · 2 the black key left of E · 3 the white key C · 4 the white key E.

TRY IT · 10 SECONDS
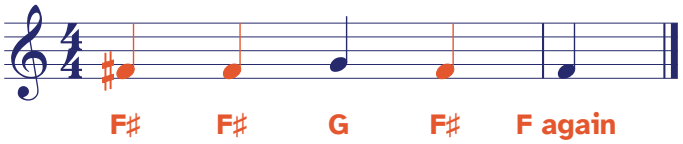


Play D, D♯, D♭.

D, then one key right (the black key), then one key left of D (another black key). Say the names as you play.

PART I · PITCH · 5 · CONTINUED

EXAMPLE 2

How long does the sharp last?

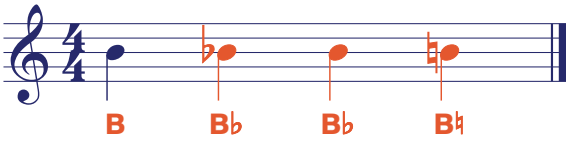


1. **The first note:** F♯.
2. **Later in the bar:** F is still F♯.
3. **After the bar line:** F is plain F again.

An accidental lasts **until the bar line**, then the key signature rules again.

EXAMPLE 3

A natural cancels the flat
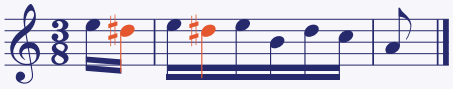


1. **B**: the white key.
2. **B♭**: one key to the left, and it stays flat for the rest of the bar.
3. **B♮**: the natural cancels the flat, back to the white key.

A natural **cancels** a sharp or flat until the bar line.

YOU KNOW THIS

Für Elise (Beethoven)



E, **D♯**, E, **D♯**: a semitone down and back. The sharp is written again in the new bar, because the bar line wiped it out.

Try it. Play E and D♯ back and forth: the black key just left of E.

THINK OF IT A key signature is the principal's notice on the school board: it speaks to every student, every day. An accidental is a teacher stopping at one student's desk: the message is for that note only, and it lasts until the next bar line.

TRAP

- × Forgetting that an accidental lasts until the bar line.
- × Carrying an accidental into the next bar: after the bar line, the key signature rules again.

SEE ALSO
Keys and key signatures p. 75

PRACTISE IT
Workbook Pitch 3 · Online: Library › Signs

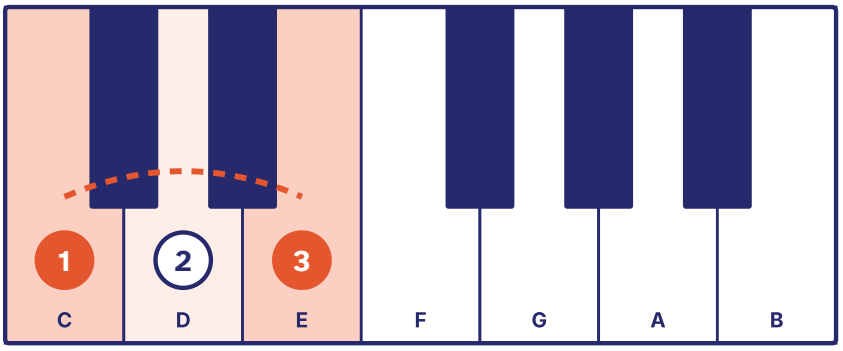
PART I · PITCH · 6

Counting intervals

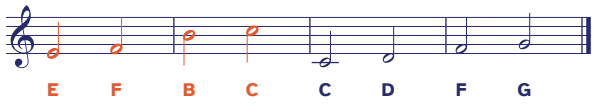
How far apart are two notes? That gap is an **interval**. The very next key is a **semitone**; two make a **tone**. To number it, count the letter names, both ends included.

THE SECRET

Count the letters, both ends included: C to E is a 3rd.



C to E: count the letters, 1 2 3, a 3rd

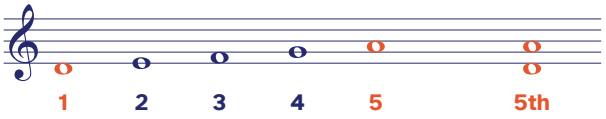


E–F and B–C are semitones; C–D and F–G are tones.



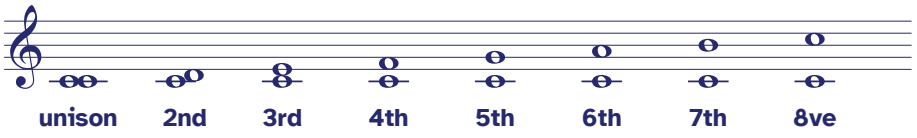
Melodic: one note after the other. Harmonic: together.

Counting the number



From **D** up to **A** (the red notes): D E F G A is five letters, so the interval is a 5th, whether the notes come one after the other or together.

Interval numbers from C

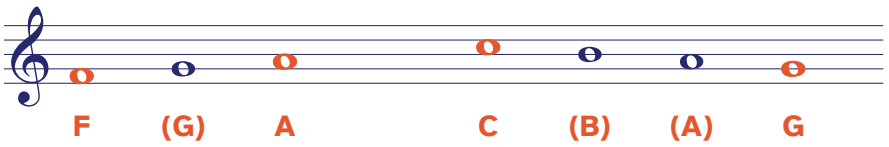


Your hand is a ruler: thumb on C, little finger on G: five fingers on five white keys, a 5th.

PART I · PITCH · 6 · CONTINUED

EXAMPLE 1

How far apart are these notes?

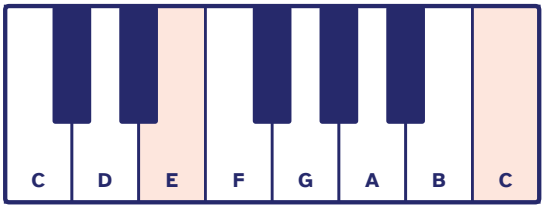


1. **Call the first note 1:** F is 1.
2. **Count every letter to the second note:** F G A is 3 letters: a 3rd.
3. **Going down works the same:** C B A G is 4 letters: a 4th.

F up to A is a **3rd**; C down to G is a **4th**. Sharps and flats never change the number.

EXAMPLE 2

Name an interval on the keyboard

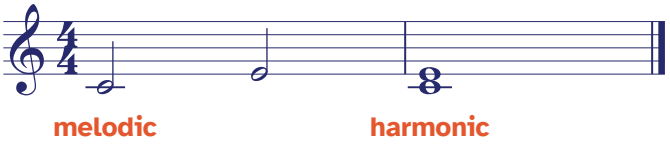


1. **Find both notes:** E and the C above it.
2. **Count the letters:** E F G A B C is six.
3. **Both ends count:** so the interval is a 6th.

E up to C is a **6th**.

EXAMPLE 3

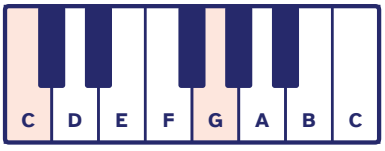
Melodic or harmonic?



1. **The first bar:** C then E, one after the other.
2. **The second bar:** C and E together.
3. **Count both:** C D E, a 3rd each time.

The same 3rd: **melodic** when played one after the other, **harmonic** when played together.

TRY IT · 10 SECONDS



Thumb on C, finger 5 on G.

Your hand is a ruler: thumb to little finger on white keys is a 5th. Play it, then thumb to finger 3: a 3rd.

PART I · PITCH · 6 · CONTINUED

GAME

Count the letters

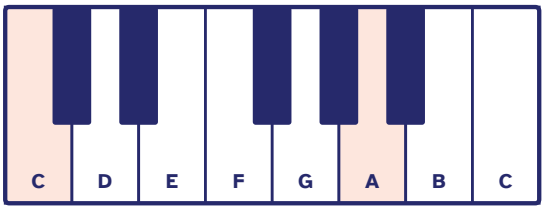
1	2	3	4
C up to G	D up to F	E up to the next E	G up to A

Count every letter, both ends included. What is each interval called?

Answer: 1 a 5th (C D E F G) · 2 a 3rd · 3 an octave (8) · 4 a 2nd.

EXAMPLE 4

Count an interval going down
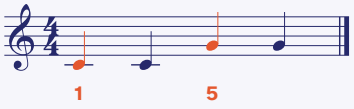


1. **Start on the top note:** A is 1.
2. **Count every letter down** to C: A G F E D C.
3. **Six letters:** a 6th.

A down to C is a **6th**, the same as C up to A.

YOU KNOW THIS

Twinkle, Twinkle, Little Star



The leap from the first “twinkle” to the second is C up to G: C D E F G, **five letters, a 5th**.

Try it. Sing the first two notes of any song, then count the letters between them.

THINK OF IT Monday to Wednesday is three days, not two: you count the day you start on. Intervals count the same way.

TRAP

- × Counting only the gaps, not the notes: always count both ends.
- × Letting a sharp or flat change the number: C to G♯ is still a 5th.

SEE ALSO

Major, minor and perfect p. 40 · Major scales p. 69

PRACTISE IT

Workbook Pitch 4 · Online: Library › Intervals

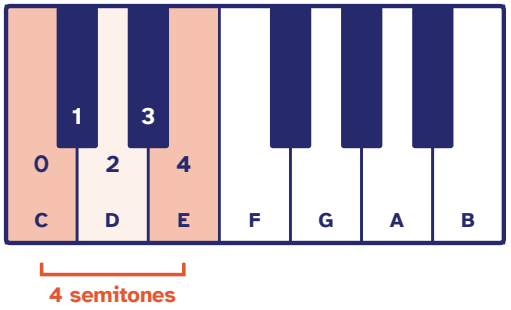
PART I · PITCH · 7

Major, minor and perfect

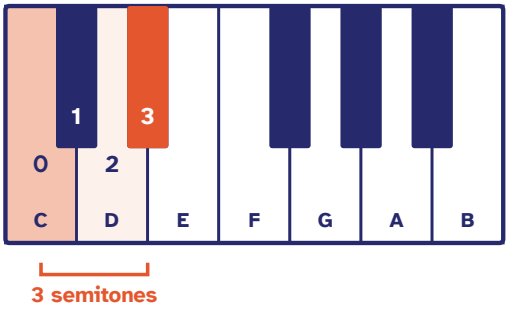
Two intervals with the same number can differ in size. C–E and C–E♭ are both 3rds, but one is a semitone wider. The **quality** (major, minor, perfect…) comes from counting **semitones**.

THE SECRET

Count the semitones: 4 make a major 3rd, 3 a minor 3rd.



From **C** to **E**: count every key, black and white. 4 semitones: a major 3rd.



From **C** to **E♭**: 3 semitones, one fewer: a minor 3rd.

One semitone less: a major 3rd becomes minor

Size in semitones

NUMBER	SMALLER	MAIN	WIDER	FIRST MET
unison		0 (perfect)		Level 1
2nd	1 (minor)	2 (major)		Level 1
3rd	3 (minor)	4 (major)		Level ½
4th		5 (perfect)	6 (augmented)	Level 1
5th	6 (diminished)	7 (perfect)		Level 1
6th	8 (minor)	9 (major)		Level 2
7th	10 (minor)	11 (major)		Level 2
octave		12 (perfect)		Level 1

2nds, 3rds, 6ths, 7ths
diminished ‹ **minor** ‹ **major** ‹ augmented

unisons, 4ths, 5ths, octaves
diminished ‹ **perfect** ‹ augmented

Each step on the ladder is one semitone. Six semitones (an augmented 4th or diminished 5th) is the tritone. Turned upside down, an interval's numbers add to 9 and major and minor swap: a major 3rd becomes a minor 6th.

PART I · PITCH · 7 · CONTINUED

EXAMPLE 1

Major or minor 3rd?



1. **Count the letters:** C D E is three, so it is a 3rd.
2. **Count the semitones:** C to C♯, D, E♭ is three.
3. **Three semitones** make a minor 3rd; four would be major.

C to E♭: a **minor 3rd**.

GAME

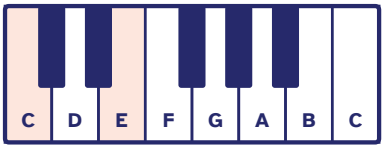
Perfect, major or minor?

1	2	3	4
C–G	D–F	C–A	E–A

Count the letters for the number, then the semitones for the size.

Answer: 1 perfect 5th (7) · 2 minor 3rd (3) · 3 major 6th (9) · 4 perfect 4th (5).

TRY IT · 10 SECONDS

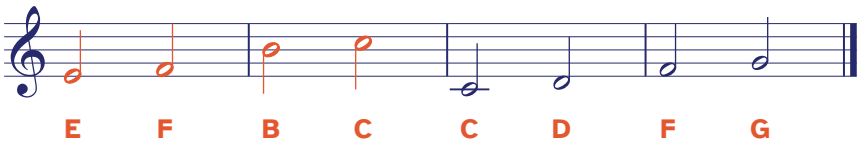


Play C–E, then C–E♭.

Bright, then dark. One key makes the difference: the major 3rd has four semitones, the minor 3rd three.

EXAMPLE 2

Tone or semitone?



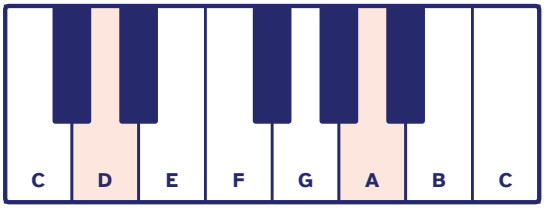
1. **E to F:** the very next key, no black key between: a semitone.
2. **C to D:** one key between (C♯): a tone.
3. **B to C** and **F to G**: check the same way.

E–F and B–C are semitones; C–D and F–G are tones.

PART I · PITCH · 7 · CONTINUED

EXAMPLE 3

Name D up to A

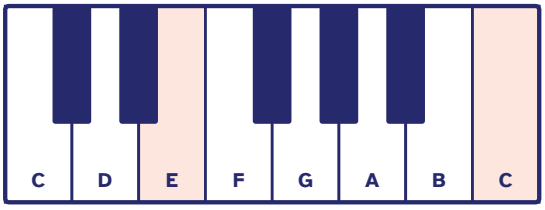


1. **The number:** D E F G A, five letters, a 5th.
2. **The size:** count every key from D to A, black and white: 7 semitones.
3. **7 semitones** is the main size of a 5th: perfect.

D up to A is a **perfect 5th**.

EXAMPLE 4

Name E up to C
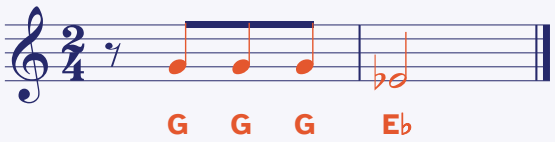


1. **The number:** E F G A B C, six letters, a 6th.
2. **The size:** 8 semitones from E to C.
3. **A major 6th** has 9, so 8 is one smaller: minor.

E up to C is a **minor 6th**.

YOU KNOW THIS

Beethoven, Symphony No. 5



Da-da-da-DUM: G down to E♭ is four semitones, a **major 3rd**. Three semitones would be minor.

Try it. Count the keys from G down to E♭: F♯, F, E, E♭.

THINK OF IT A major and a minor 3rd are a smile and a frown: the same face, one small change at the corners.

TRAP

- × Counting white keys only: count every key, black and white.
- × Calling a 4th or 5th “major”: they are perfect.

SEE ALSO

Counting intervals p. 37 · Chords and triads p. 86

PRACTISE IT

Workbook Pitch 4 · Online: Library › Intervals

PART I · PITCH · TIPS

When a note reads wrong

Most misread notes come from a handful of slips. Name the slip first, then fix only that note: playing the whole line again rarely finds it.

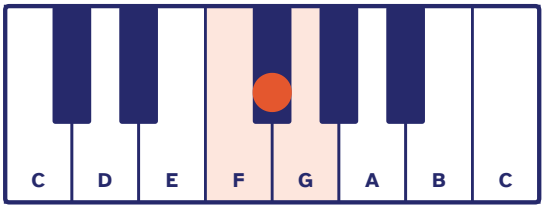
TIP

What you notice, what to check

WHAT YOU NOTICE	CHECK	TURN TO
The left hand reads like the right	**the clef**: bass lines are G B D F A, not E G B D F	p. 25
You land an octave too high or too low	**a landmark**: middle C, treble G or bass F	p. 28
Notes above or below the staff slow you down	**the ledger lines**: count them from the nearest landmark	p. 28
A sharp or flat vanishes, or stays too long	**the bar line**: an accidental lasts only to the end of its bar	p. 34
You play the black key on the wrong side	**the direction**: a sharp goes right (up), a flat goes left (down)	p. 34
Two notes sound too close or too far apart	**the interval**: count the letters, then the keys between	p. 37

EXAMPLE

Same key, two names



1. **From F, go one key right:** the red black key is **F♯**.
2. **From G, go one key left:** you reach the same red key, **G♭**.
3. **One key, two names:** the music decides which name it uses.

If F♯ and G♭ both look odd, check the key: the name follows the notes around it, the key stays the same.

HOW TO

Three checks before you read a new line

1. **The clefs:** which staff is treble, which is bass?
2. **The landmarks:** find middle C, treble G and bass F, and read the other notes from them.
3. **The accidentals:** mark each sharp, flat or natural, and see where its bar ends.

THINK OF IT Like reading a map: find the big landmarks first, then the streets around them. A note far from any landmark is the one most often misread.